The Fabius Function is Not an Elementary Function

Density of the analytic locus, and what it costs to prove

Human-readable account of the formal development
Vladimir Reshetnikov's `ProveIt` repository

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Abstract

The Fabius function F is a C^∞ probability distribution function on $[0, 1]$ that is real analytic at no point of that interval. We prove, and formalize in Lean 4 on top of `Mathlib`, that no elementary function of one real variable agrees with F on any nonempty open subset of $[0, 1]$; in particular the restriction of F to $(0, 1)$ is not elementary.

Since F is C^∞, no smoothness obstruction can decide the question: the obstruction must be analyticity. The whole content of the argument is therefore a single positive statement about the class of elementary functions, which we prove in the form we need it and no stronger:

the set of points at which an elementary function is real analytic is a dense open subset of $\mathbb{R}$.

Density is exactly right and cannot be improved: $x \mapsto x^{-1}$ and $x \mapsto |x|$ are elementary and fail to be analytic at the origin. The induction that proves it is short but has one genuinely delicate step, the unary one, where a locally constant intermediate value must be separated from a locally nonconstant one; the separation is furnished by the isolated-zeros theorem for real analytic functions of one variable.

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1 Statement

1.1 The two inputs

Write $\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}$ for the *analytic locus* of $f\colon \mathbb{R} \to \mathbb{R}$. The proof combines one negative fact about the Fabius function with one positive fact about elementary functions.

Theorem 1.1 (Nowhere analyticity; already formalized)**.** *Let F be the Fabius function, extended by 0 to the left of 0 and by 1 to the right of 1. Then $\mathcal{A}(F) = \mathbb{R} \setminus [0, 1]$. In particular F is real analytic at no point of $[0, 1]$, endpoints included.*

This is the unnumbered corollary to equation (24) of Arias de Reyna, *Arithmetic of the Fabius function* (arXiv:1702.06487v3), transferred from Rvachev's up to F and completed at the right endpoint; in the repository it is `Fabius.canonical_fabius_analyticAt_iff` in the module `FabiusFunction.NowhereAnalytic`. Nothing in the present document reproves it.

Theorem 1.2 (Density of the analytic locus)**.** *If $f\colon \mathbb{R} \to \mathbb{R}$ is elementary, then $\mathcal{A}(f)$ is a dense open subset of $\mathbb{R}$.*

Corollary 1.3 (Main result)**.** *Let $U \subseteq [0, 1]$ be a nonempty open set and let $g\colon \mathbb{R} \to \mathbb{R}$ be elementary. Then g and F do not agree on U. In particular there is no elementary function that agrees with F on $(0, 1)$, and F itself is not elementary.*

Proof. By Theorem 1.2 the set $\mathcal{A}(g)$ is dense, so it meets the nonempty open set U; pick $x \in U \cap \mathcal{A}(g)$. If g and F agreed on U they would have the same germ at x, because U is a neighbourhood of x; analyticity at a point depends only on the germ, so F would be analytic at $x \in [0, 1]$, contradicting Theorem 1.1. □

Theorem 1.3 is the sharp form of the assertion "the Fabius function is not elementary", and it is strictly stronger than the bare non-elementarity of F as a function on $\mathbb{R}$. The bare statement leaves open the possibility that some elementary function agrees with F on all of $(0, 1)$ and differs from it outside; Theorem 1.3 rules that out, and rules it out on every subinterval.

Remark 1.4. Only a very weak consequence of Theorem 1.2 is actually used: that an elementary function is analytic at *at least one* point of every nonempty open set. Density is nevertheless the statement one must prove, because it is the statement that survives the induction: "analytic somewhere" is not preserved by intersection, whereas "dense open" is.

2 The class of elementary functions

2.1 Definition

We follow the description at `https://en.wikipedia.org/wiki/Elementary_function`: elementary functions are those obtained from polynomials, roots, exponentials, logarithms, and the trigonometric functions and their inverses by finitely many applications of the field operations and composition. Formally:

Definition 2.1 (Elementary functions)**.** The class $\mathcal{E}$ of *elementary functions* $\mathbb{R} \to \mathbb{R}$ is the smallest class of functions such that

1. every constant function $x \mapsto c$ lies in $\mathcal{E}$, and the identity $x \mapsto x$ lies in $\mathcal{E}$;
2. if $f, g \in \mathcal{E}$ then $x \mapsto f(x) + g(x)$, $x \mapsto f(x)g(x)$ and $x \mapsto -f(x)$ lie in $\mathcal{E}$;
3. if $f \in \mathcal{E}$ then $x \mapsto f(x)^{-1}$ lies in $\mathcal{E}$;
4. if $f \in \mathcal{E}$ and $r \in \mathbb{R}$ then $x \mapsto f(x)^r$ lies in $\mathcal{E}$;
5. if $f \in \mathcal{E}$ then each of $\exp \circ f$, $\log \circ f$, $\sin \circ f$, $\cos \circ f$, $\arcsin \circ f$, $\arctan \circ f$ lies in $\mathcal{E}$.

Two features of Theorem 2.1 deserve comment.

Composition is a theorem, not an axiom. Theorem 2.1 contains no clause "$f, g \in \mathcal{E} \Rightarrow f \circ g \in \mathcal{E}$". It does not need one.

Proposition 2.2. *$\mathcal{E}$ is closed under composition.*

Proof. Induct on the derivation of the outer function f. Every clause of Theorem 2.1 commutes with precomposition by a fixed g: the constant $x \mapsto c$ composed with g is again that constant; the identity composed with g is g itself, which is elementary by hypothesis; and $(f_1 + f_2) \circ g = (f_1 \circ g) + (f_2 \circ g)$, $(\exp \circ f) \circ g = \exp \circ (f \circ g)$, and likewise for every remaining clause. □

Consequently $\mathcal{E}$ contains every function one expects: differences $f - g = f + (-g)$, quotients $f/g = f \cdot g^{-1}$, natural powers, all polynomial and rational functions, $\sqrt{\cdot}$ and all n-th roots $x \mapsto x^{1/n}$, the absolute value $|x| = \sqrt{x^2}$, the tangent $\sin / \cos$, the hyperbolic functions $\sinh x = (\mathrm{e}^x - \mathrm{e}^{-x})/2$, $\cosh$, $\tanh$, the arccosine $\pi/2 - \arcsin$, and the inverse hyperbolic sine $\operatorname{arsinh} x = \log\left(x + \sqrt{1 + x^2}\right)$.

Total functions and junk values. Every member of $\mathcal{E}$ is a *total* function $\mathbb{R} \to \mathbb{R}$. This is a deliberate choice, and it is the choice that makes Theorem 1.3 strongest. The formalization inherits `Mathlib`'s conventions

$$0^{-1} = 0, \qquad \log x = \log|x|, \qquad \log 0 = 0, \qquad x^r = |x|^r \cos(\pi r) \ (x < 0), \qquad \arcsin x = \pm\tfrac{\pi}{2} \ (\pm x \geq 1),$$

so that clauses (3)–(5) of Theorem 2.1 never fail to produce a function.

One might worry that junk values weaken the theorem, by admitting "elementary" functions that are not classically elementary. They do the opposite. Admitting more functions into $\mathcal{E}$ makes the assertion "$g \in \mathcal{E}$ and $g = F$ on U" easier to satisfy, hence its refutation stronger. And nothing is lost in the other direction: if a classical elementary expression is well formed at every point of an open set U — no vanishing denominator, no nonpositive logarithm, no arcsine of a value outside $[-1, 1]$ — then on U it agrees pointwise with the total function produced by the same derivation, since on U the junk values are never consulted. So every classical elementary function on U is the restriction to U of a member of $\mathcal{E}$, and Theorem 1.3 applies to it.

2.2 What is not claimed

$\mathcal{E}$ is closed under n-th roots but not, as defined, under passage to an arbitrary algebraic function: Theorem 2.1 has no clause producing a continuous branch of $P(x, y) = 0$ for a polynomial P with elementary coefficients whose Galois group is not solvable. Liouville's differential field definition of "elementary" does admit such branches. We do not claim them here. The obstruction is purely one of formalization cost: the analogue of Theorem 1.2 for algebraic branches is true and classical — away from the zero set of the discriminant, the analytic implicit function theorem applies, and that zero set is finite on each interval — but the analytic implicit function theorem over $\mathbb{R}$ is not available in `Mathlib` at the time of writing.

We also make no claim about elementary functions of a complex variable, and none about the closely related but distinct question of whether F satisfies an algebraic differential equation.

3 Analyticity of the generators

Each generator of Theorem 2.1 is real analytic away from a finite set of arguments. All of these facts are in `Mathlib`, most of them in the form "C^n for every n in $\mathbb{N}\cup\{\infty,\omega\}$"; specializing to the analytic exponent ω and applying `ContDiffAt.analyticAt` converts them to the statements below.

Lemma 3.1.

1. $\exp$, $\sin$, $\cos$ *and* $\arctan$ *are real analytic at every point of* $\mathbb{R}$.
2. $\log$ *is real analytic at every* $x \neq 0$ *— at negative arguments as well, because* $\log$ *is even under the convention* $\log x = \log|x|$.
3. *For each fixed* $r \in \mathbb{R}$*, the function* $x \mapsto x^r$ *is real analytic at every* $x \neq 0$*; at negative* x *this uses the identity* $x^r = |x|^r \cos(\pi r)$.
4. $\arcsin$ *is real analytic at every* $x \notin \{-1, 1\}$*; outside* $[-1, 1]$ *it is locally constant.*
5. $x \mapsto x^{-1}$ *is real analytic at every* $x \neq 0$.

Remark 3.2. The bad sets are sets of *values* of the inner function, not sets of points of the domain — $\{0\}$ for the reciprocal, the logarithm and a real power, $\{-1, 1\}$ for the arcsine, and $\varnothing$ for the rest. This is the shape the induction consumes, and it is why Theorem 4.2 below is stated with a finite set $B \subseteq \mathbb{R}$ of forbidden values.

4 Density of the analytic locus

4.1 Openness

Lemma 4.1. *For every* $f : \mathbb{R} \to \mathbb{R}$ *the set* $\mathcal{A}(f)$ *is open.*

Proof. Analyticity at x means that f agrees near x with the sum of a convergent power series; that series then converges on a whole ball around x and, after a change of origin, represents f near every point of that ball. So $\mathcal{A}(f)$ contains a neighbourhood of each of its points. □

4.2 The unary step

The following lemma is where the argument is actually made. Recall the shape of the difficulty. If f is analytic on a dense open set and g is analytic everywhere except at finitely many values, one would like to say that $g \circ f$ is analytic wherever f is analytic and $f(x)$ avoids the bad values. That set need not be dense: consider $f \equiv 0$ and $g(y) = y^{-1}$. What saves the statement is that in that degenerate case $g \circ f$ is constant, hence analytic anyway. Separating the two cases is exactly the isolated-zeros theorem.

Lemma 4.2 (Key lemma)**.** *Let* $f, g : \mathbb{R} \to \mathbb{R}$ *and let* $B \subseteq \mathbb{R}$ *be finite. Suppose* $\mathcal{A}(f)$ *is dense and* g *is analytic at every* $y \notin B$*. Then* $\mathcal{A}(g \circ f)$ *is dense.*

Proof. Let U be a nonempty open set; we must produce a point of $U \cap \mathcal{A}(g \circ f)$. Since $\mathcal{A}(f)$ is dense there is $x_0 \in U \cap \mathcal{A}(f)$, and by Theorem 4.1 the set $V = U \cap \mathcal{A}(f)$ is an open neighbourhood of x_0. Distinguish two cases.

Case 1: f is constant equal to some $c \in B$ on a neighbourhood of x_0. Then $g \circ f$ equals the constant $g(c)$ on that neighbourhood, so $g \circ f$ is analytic at x_0 itself, and $x_0 \in U$.

Case 2: for no $c \in B$ is f eventually equal to c near x_0. Fix $c \in B$. The function $f - c$ is analytic at x_0, so by the isolated-zeros theorem for analytic functions of one real variable, either $f - c$ vanishes identically near x_0 — excluded in this case — or $f - c$ is nonvanishing on some punctured neighbourhood of x_0. Thus for each of the finitely many $c \in B$ there is a punctured neighbourhood of x_0 on which $f \neq c$. Intersecting these finitely many punctured neighbourhoods with the punctured neighbourhood $V \setminus \{x_0\}$ leaves a punctured neighbourhood of x_0, which is nonempty because $\mathbb{R}$ has no isolated points. Any point t of it satisfies $t \in U$, f analytic at t, and $f(t) \notin B$; hence g is analytic at $f(t)$ and $g \circ f$ is analytic at t. □

Remark 4.3. The case distinction is not an artefact of the write-up. Without it the lemma is false as stated for the naive reason above, and with a merely C^∞ inner function it is false outright: analyticity of f is what makes the dichotomy "locally constant or locally injective away from a discrete set" available. This is the only place in the proof where a theorem of substance about real analytic functions is used.

4.3 The induction

Proof of Theorem 1.2. Openness of $\mathcal{A}(f)$ is Theorem 4.1 and holds for every f. Density is proved by induction on the derivation of f in Theorem 2.1.

Constants and the identity. Here $\mathcal{A}(f) = \mathbb{R}$.

Sums and products. If $\mathcal{A}(f)$ and $\mathcal{A}(g)$ are dense then so is $\mathcal{A}(f) \cap \mathcal{A}(g)$, because a finite intersection of dense open sets is dense. Analyticity is preserved by sums and products pointwise, so $\mathcal{A}(f) \cap \mathcal{A}(g) \subseteq \mathcal{A}(f+g)$ and likewise for fg; a superset of a dense set is dense.

Negation. $\mathcal{A}(f) \subseteq \mathcal{A}(-f)$.

The unary clauses. Apply Theorem 4.2 with the pair (f, g) and the finite bad set B read off from Theorem 3.1: $B = \varnothing$ for exp, sin, cos, arctan; $B = \{0\}$ for $y \mapsto y^{-1}$, for log, and for $y \mapsto y^r$; and $B = \{-1, 1\}$ for arcsin. □

Remark 4.4 (Sharpness). Density cannot be strengthened to "$\mathcal{A}(f) = \mathbb{R}$": the elementary functions $x \mapsto x^{-1}$, $x \mapsto |x|$ and $x \mapsto \arcsin x$ have analytic loci $\mathbb{R} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$ and $\mathbb{R} \setminus \{-1, 1\}$. Nor can "dense open" be strengthened to "open with finite complement": the elementary function $x \mapsto (\sin x)^{-1}$ omits the infinite discrete set $\pi\mathbb{Z}$. What density does give for free is that the complement is closed and nowhere dense, being the complement of a dense open set. Whether the complement of the analytic locus of an elementary function is always *discrete* is a natural further question that we neither prove nor use.

5 The formal development

5.1 Correspondence with the Lean names

All statements below live in the namespace `Fabius`.

Object or statement	Lean name	Module
Elementary functions (Theorem 2.1)	`IsElementary`	`ElementaryFunction`
Closure under composition (Theorem 2.2)	`IsElementary.comp`	`ElementaryFunction`
Derived closures: $-$, $/$, x^n, $\sqrt{\cdot}$, $\lvert\cdot\rvert$, tan, sinh, cosh, tanh, arccos, arsinh, polynomials	`IsElementary.sub`, `.div`, `.npow`, `.sqrt`, `.abs`, `.tan`, `.sinh`, `.cosh`, `.tanh`, `.arccos`, `.arsinh`, `.polynomial`	`ElementaryFunction`
Analytic locus $\mathcal{A}(f)$	`analyticLocus`	`ElementaryFunction`
Openness (Theorem 4.1)	`isOpen_analyticLocus`	`ElementaryFunction`
Generators (Theorem 3.1)	`analyticAt_log`, `analyticAt_rpow_const`, `analyticAt_sqrt`, `analyticAt_arctan`, `analyticAt_arcsin`	`ElementaryFunction`
Key lemma (Theorem 4.2)	`dense_analyticLocus_comp`	`ElementaryFunction`
Density (Theorem 1.2)	`IsElementary.dense_analyticLocus`	`ElementaryFunction`
Analytic at a point of any nonempty open set	`IsElementary.exists_analyticAt_of_isOpen`	`ElementaryFunction`
Nowhere analyticity (Theorem 1.1)	`canonical_fabius_not_analyticAt`, `canonical_fabius_analyticAt_iff`	`NowhereAnalytic`
Main result (Theorem 1.3)	`canonical_fabius_not_isElementary_on_Ioo`	`NotElementary`
F is not elementary	`canonical_fabius_not_isElementary`	`NotElementary`
General open subset of $[0, 1]$	`not_isElementary_eqOn`	`NotElementary`
Rvachev's up and the signed extension	`rvachev_not_isElementary`, `extendedFabius_not_isElementary`	`NotElementary`

All modules are under `Analysis/FabiusFunction/Lean/FabiusFunction/`.

5.2 Ingredients taken from `Mathlib`

The formalization is short because `Mathlib` already contains the two theorems of substance that the argument consumes.

- `AnalyticAt.eventually_eq_zero_or_eventually_ne_zero` and its two-function form: the isolated-zeros theorem for a function of one variable over a nontrivially normed field, which is what powers Theorem 4.2. It applies verbatim to $f : \mathbb{R} \to \mathbb{R}$.
- `AnalyticAt.eventually_analyticAt`: analyticity at a point propagates to a neighbourhood, which is Theorem 4.1.
- `ContDiffAt.analyticAt` together with the family of `contDiffAt_. . .` lemmas stated for exponents in $\mathbb{N} \cup \{\infty, \omega\}$: this is how Theorem 3.1 is obtained, including the two facts that are otherwise awkward over $\mathbb{R}$, namely analyticity of $\arctan$ everywhere and of $\arcsin$ off $\{\pm 1\}$.

5.3 Statement of the main theorem in Lean

```
theorem canonical_fabius_not_isElementary_on_Ioo :
    not (exists g : R -> R, IsElementary g and
         Set.EqOn g (fabiusReal fabius) (Set.Ioo 0 1))
```

(rendered here in ASCII; the source uses the usual notation). The development contains no `sorry`, no `axiom` and no `opaque`, and its axiom set is the `Mathlib` default `propext`, `Classical.choice`, `Quot.sound`.

6 Discussion

6.1 Why smoothness cannot decide the question

The Fabius function is C^∞ on $\mathbb{R}$, and its derivatives satisfy the exact self-similar relation $F'(x) = 2F(2x)$ on $[0, 1/2]$. Any proof of non-elementarity must therefore distinguish F from the elementary functions by a property strictly finer than differentiability of every order. Real analyticity is the natural candidate, and Theorem 1.2 says it is a sufficient one. It is also, in a precise sense, the *only* such candidate we know how to extract from Theorem 2.1 directly. Nothing in this document rules out a different separating property; what it does show is that analyticity suffices, and that it is available at the modest cost of Theorem 4.2.

6.2 The role of the endpoints

Theorem 1.1 includes the endpoints 0 and 1, where F is flat to infinite order; this is what makes Theorem 1.3 apply to *every* nonempty open subset of $[0, 1]$ rather than to open subsets of $(0, 1)$ only. Note that $\mathcal{A}(F)$ is exactly $\mathbb{R} \setminus [0, 1]$: outside the unit interval F is locally constant, hence analytic, so the nowhere-analytic locus is as large as it can be and no larger.

6.3 What a stronger theorem would require

Two natural strengthenings are visible from here.

1. *Liouvillian functions.* Replacing $\mathcal{E}$ by the class of functions lying in a Liouville tower over the rational functions — allowing arbitrary algebraic branches, and iterated exponentials and logarithms in the differential-field sense — would subsume Theorem 2.1. The analytic content is unchanged; what is missing is the analytic implicit function theorem over $\mathbb{R}$, as recorded in Section 2.2.

2. *Differential algebraicity.* A stronger and genuinely different statement is that F satisfies no algebraic differential equation $P\big(x, F, F', \ldots, F^{(n)}\big) = 0$ with polynomial P. Such a statement does not follow from nowhere analyticity — differentially algebraic functions need not be analytic — and is not addressed here.